



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

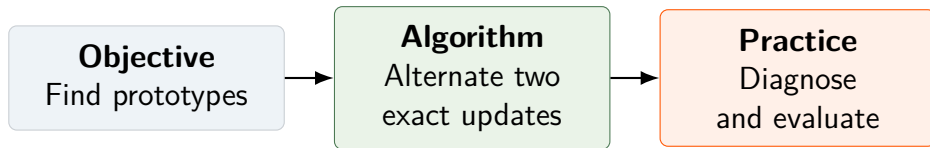
COMP 5212

Machine Learning

Instructor: Zihan Zhang

Lecture 8: K-means Clustering

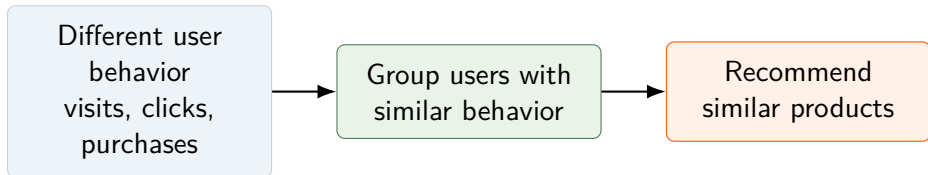
Learning Goals



- Justify squared error and derive the two Lloyd updates;
- Distinguish local-optimum facts from approximation guarantees;
- Choose preprocessing, K , and evaluation checks;
- Use a Johnson–Lindenstrauss projection to reduce dimension

Motivation and Objective

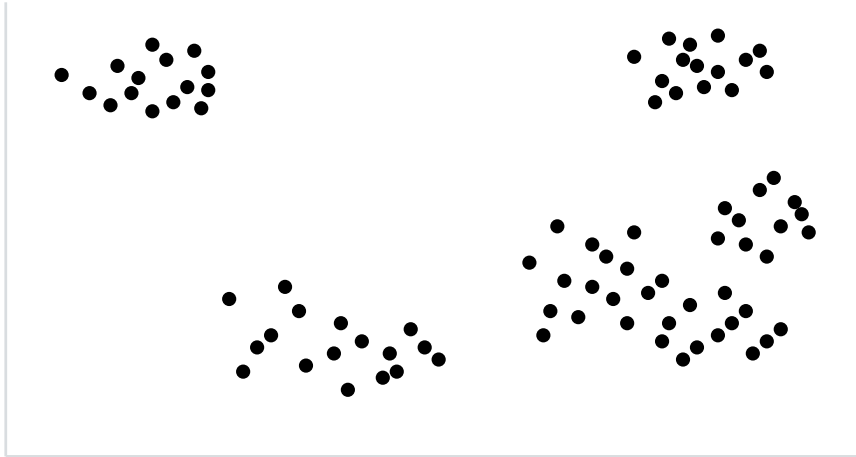
User-Behavior Clustering



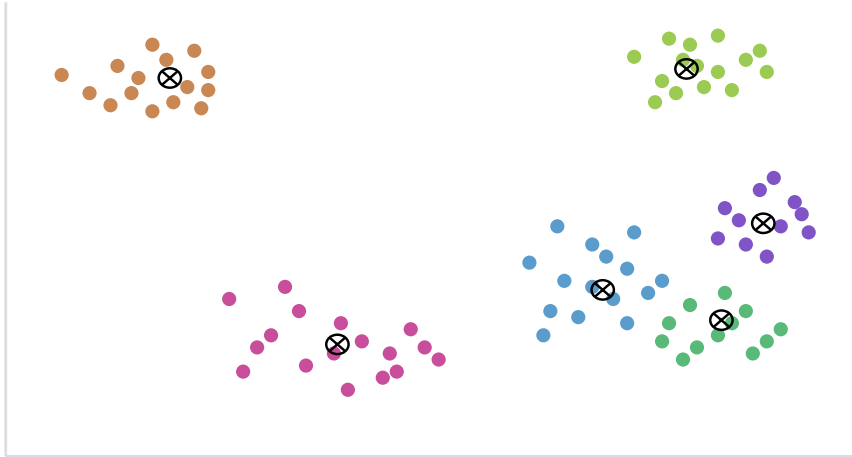
Users with similar behavior may prefer similar products, so behavior clusters can support recommendations and revenue.

- Why not classification?

A Clustering Example



A Clustering Example



What Is the Object We Want to Learn?

Observed data

$$x_1, \dots, x_n \in \mathbb{R}^d.$$

For each proposed cluster C_k ,
compute its coordinate-wise mean

$$\bar{x}_{C_k} = \frac{1}{|C_k|} \sum_{i \in C_k} x_i.$$

Unknown structure

$C_1, \dots, C_K$ a partition of $\{1, \dots, n\}$.

- Labels are not supplied;
- K is chosen before fitting;
- compactness is measured in the selected feature geometry.

Goal: find a partition whose members are close to their own cluster mean.

Why Use Squared Error?

For a partition $C_1, \dots, C_K$, define the empirical within-cluster error directly from the means:

$$\text{SSE}(C_1, \dots, C_K) = \sum_{k=1}^K \sum_{i \in C_k} \|x_i - \bar{x}_{C_k}\|^2.$$

- Robust to general noise assumptions;
- Feasible to dimension reductions.

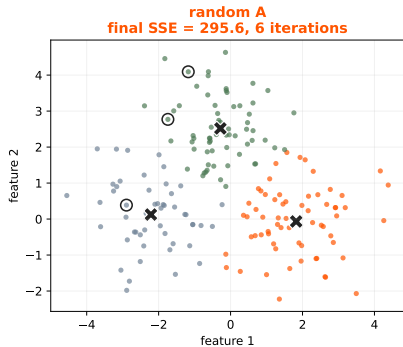
The Algorithm

Lloyd's Algorithm

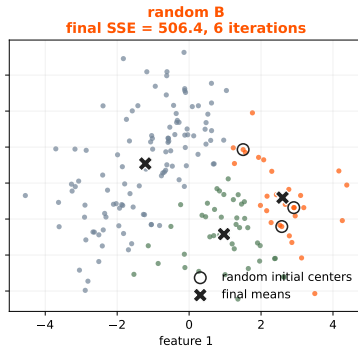
1. Choose K initial centers $\mu_1, \dots, \mu_K$.
2. **Assignment:** $z_i \leftarrow \arg \min_k \|x_i - \mu_k\|^2$.
3. **Update:** $\mu_k \leftarrow \frac{1}{|C_k|} \sum_{i: z_i=k} x_i$.
4. Repeat until assignments, centers, or J stop changing.

A 2-D K-means Demo in the Browser

Same 2-D data, independent random Lloyd starts

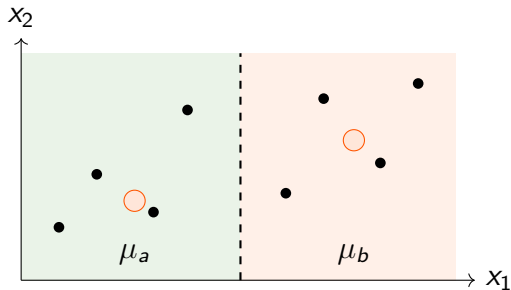


Synthetic blobs · $K = 3$ · hollow circles are random initial centers · X markers are final means



○ random initial centers
X final means

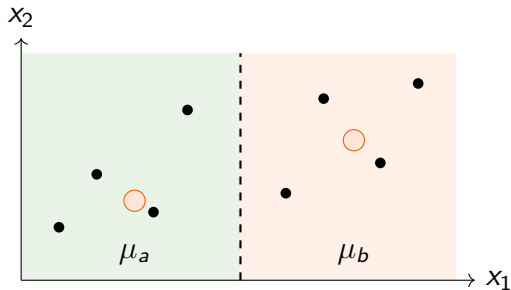
The Assignment Step Creates Voronoi Cells



A point belongs to the center with the smaller squared distance.

$$\|x - \mu_a\|^2 \leq \|x - \mu_b\|^2.$$

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The boundary is a hyperplane; K-means **cannot** express arbitrary non-convex cells.

Why Does Lloyd's Objective Decrease?

Assignment step

$$J(z^{t+1}, \mu^t) \leq J(z^t, \mu^t).$$

Each sample chooses its nearest current center.

Update step

$$J(z^{t+1}, \mu^{t+1}) \leq J(z^{t+1}, \mu^t).$$

Each center becomes the best mean for its current cluster.

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$$J^0 \geq J^1 \geq J^2 \geq \dots \geq 0.$$

Monotone descent gives a fixed point, not a guarantee of the global optimum.

Theory and Practice

What Does a Lloyd Fixed Point Guarantee?

For arbitrary initialization

- With deterministic tie-breaking, SSE is non-increasing and the finite assignment set leads to a fixed point.
- A fixed point is stationary under the two Lloyd updates, not globally optimal.
- Arbitrary Lloyd starts have **no** finite worst-case approximation ratio.

What initialization can add

- k-means++ [AV2007] gives

$$\mathbb{E}[J_{\text{init}}] \leq 8(\ln K + 2) J^*.$$

- Lloyd iterations only decrease ϕ , so the same expected upper bound holds for the final run.
- Computing J^* is NP-hard in the general case.

[AV2007] k-means++: The Advantages of Careful Seeding. David Arthur and Sergei Vassilvitskii, SODA 2007, pp. 1027–1035.

Initialization Is Part of the Algorithm

Random starts

- Simple, but centers may start together.
- A single seed can be misleading.

k-means++

- Choose the first center at random.
- Sample later centers with probability proportional to squared distance from the current set.

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In practice: use k-means++, several restarts, a fixed seed for reproducibility, and report the spread of final objectives.

Implementation Details Change the Result

- Standardize or transform features before measuring distances.
- Decide how to handle missing values and empty clusters.
- Remove or robustly treat influential outliers.
- Use MiniBatchKMeans when one full pass is too expensive.

How Should We Choose K ?

Inertia

$$\text{inertia}(K) = \min_{z, \mu} J(z, \mu).$$

It never increases as K grows.

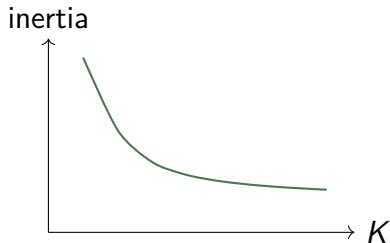
Silhouette

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}.$$

Compare within-cluster and nearest-other-cluster distances.

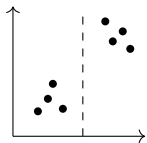
- a_i : the average distance to the datapoints within the same clique
 $\frac{1}{|C_i|-1} \sum_{j \in C_i, j \neq i} \|x_i - x_j\|$
- b_i : the minimal average distance to a neighbor clique $\min_{C \neq C_i} \frac{1}{|C|} \sum_{j \in C} \|x_i - x_j\|$.

A Low Inertia Is Not Always Good



- Increasing K always improves the training objective.
- A stable cluster across seeds or resamples is more credible.
- A useful cluster must have an interpretable consequence.

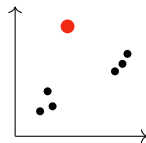
Failure Gallery: The Geometry Is the Model



Well-separated blobs



Non-convex structure



Unequal density and outliers

When K-means fails, ask whether the distance, scale, K , or cluster shape is mismatched to the data.

Complexity and Scaling to Larger Data

Total computational cost $O(nKdt)$.

- n : the size of dataset;
- K : the number of clusters;
- d : feature dimension;
- t : number of iteration steps.

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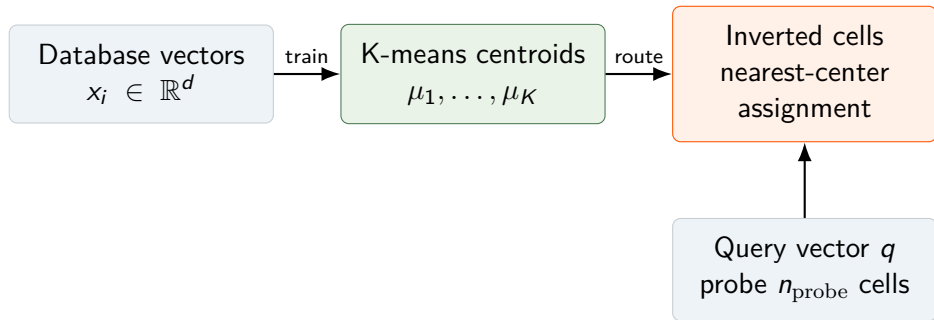
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Approaches for scaling up

- Mini-batches reduce the cost of each update.
- Sparse or approximate distance routines reduce memory pressure.

Applications

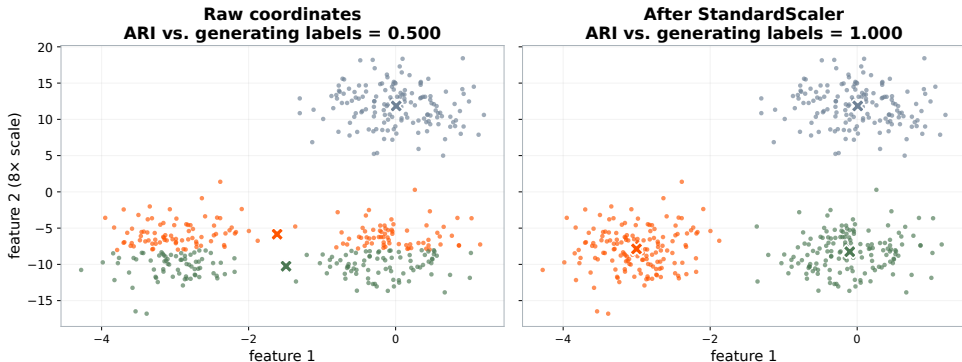
Application: K-means as an IVF Search Coarse Quantizer



- x_i is a stored embedding or feature vector; a cluster is one coarse inverted cell.
- A query probes selected cells, then exact distances rerank candidates; n_{probe} trades recall against latency.

Experiments Expose the Preprocessing Assumptions

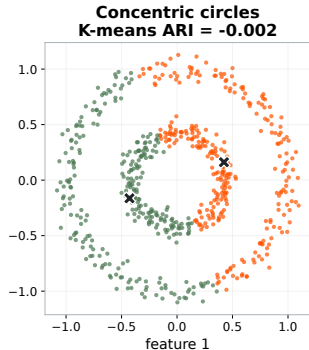
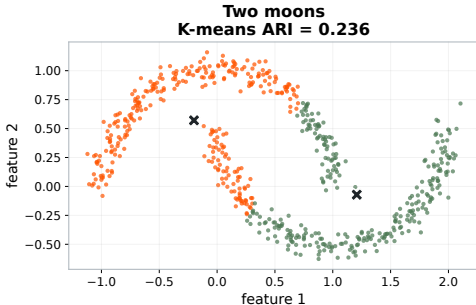
Scale changes the Euclidean geometry



Synthetic data · identical original coordinates in both panels · X markers are fitted centroids

Experiments Expose the Geometry Assumptions

A Voronoi partition cannot follow every shape



Synthetic data · colors are K-means assignments (n_clusters = 2, n_init = 10, seed = 0)

Remedies for the Non-regular Case

Change the representation

- Engineer a radial feature, or use kernel K-means as an implicit nonlinear map.
- For concentric rings, $u = x_1^2 + x_2^2$ is a radius coordinate.
- K-means on u uses a one-dimensional threshold.

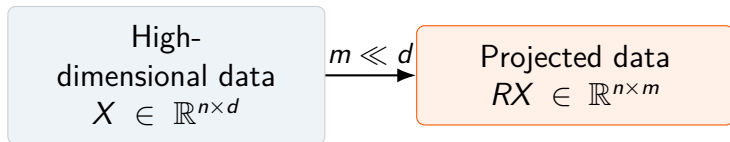
Challenge: standard K-means does not train a feature map; the useful feature must come from data geometry or domain knowledge.

A Practical Reporting Checklist

1. State what one row x represents and which features were used.
2. Record scaling, transformations, missing-value handling, and distance.
3. Run multiple initializations and report objective variability.
4. Compare several K values using internal and stability diagnostics.
5. Inspect cluster sizes, profiles, and outliers.
6. Keep labels or downstream outcomes for validation, not for fitting.

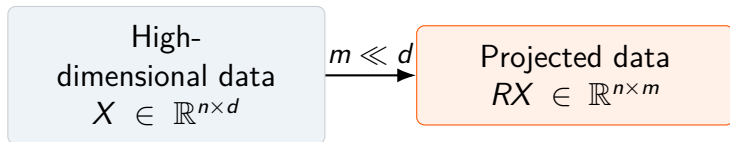
Distance-Preserving Reduction

Random Projection Before K-means



- Lower memory and distance-computation cost;
- useful when d is much larger than $\log n$.
- Preserve sample geometry;
- projected coordinates need not be interpretable.

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The question is whether the K-means objective survives the projection.

Johnson–Lindenstrauss Lemma

For n points and failure probability δ , a random linear map $R : \mathbb{R}^d \rightarrow \mathbb{R}^m$ with

$$m = O\left(\frac{\log(n/\delta)}{\varepsilon^2}\right)$$

A common Gaussian-projection implementation uses the sufficient bound

$$m \geq \frac{4 \log n}{\varepsilon^2/2 - \varepsilon^3/3}.$$

can satisfy, simultaneously for every pair i, j ,

$$(1 - \varepsilon) \|x_i - x_j\|^2 \leq \|Rx_i - Rx_j\|^2 \leq (1 + \varepsilon) \|x_i - x_j\|^2.$$

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Typical choices: Gaussian entries scaled by $1/\sqrt{m}$,
random signs, or a sparse random projection.

JL Preserve a K-means Cluster Cost

For a cluster C with mean μ_C ,

$$\sum_{i \in C} \|x_i - \mu_C\|^2 = \frac{1}{2|C|} \sum_{i,j \in C} \|x_i - x_j\|^2.$$

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If JL preserves all pairwise distances, it preserves every fixed clustering cost within $1 \pm \varepsilon$.

The Projected Optimum Is Approximately Competitive

Let C^* minimize the original K-means objective and $\tilde{C}$ minimize the projected objective. For every partition C ,

$$(1 - \varepsilon)J(C) \leq J_R(C) \leq (1 + \varepsilon)J(C).$$

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$$J(\tilde{C}) \leq \frac{J_R(\tilde{C})}{1 - \varepsilon} \leq \frac{J_R(C^*)}{1 - \varepsilon} \leq \frac{1 + \varepsilon}{1 - \varepsilon} J(C^*).$$

K-means in One Sentence

K-means learns a finite set of prototypes by alternating nearest-center assignment and mean updates under a chosen Euclidean geometry.

- The objective explains the updates;
- the geometry explains the failure modes;
- initialization and preprocessing shape the result;
- evaluation must include stability and domain meaning;
- JL projection can reduce dimension while approximately preserving the objective.